E102 Motion of Celestial Bodies under Gravitation

E1.2.1 Kepler's first law

A planet follows an ellipse with the Sun at one focus, not at its centre. Its Sun-planet distance therefore changes along the orbit. A circle is the special case with coincident foci. Distinguish major axis, semi-major axis and instantaneous distance; they are not one interchangeable radius.

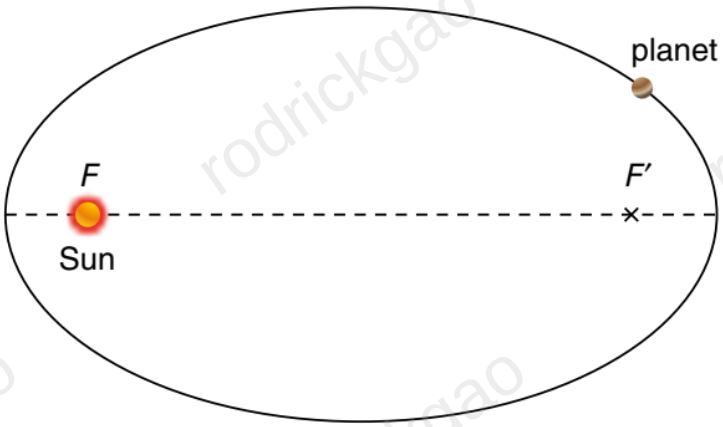


Fig 2.1a *Kepler's first law.*

The Sun lies at one focus F. The other focus is empty; the orbit is not centred on the Sun.

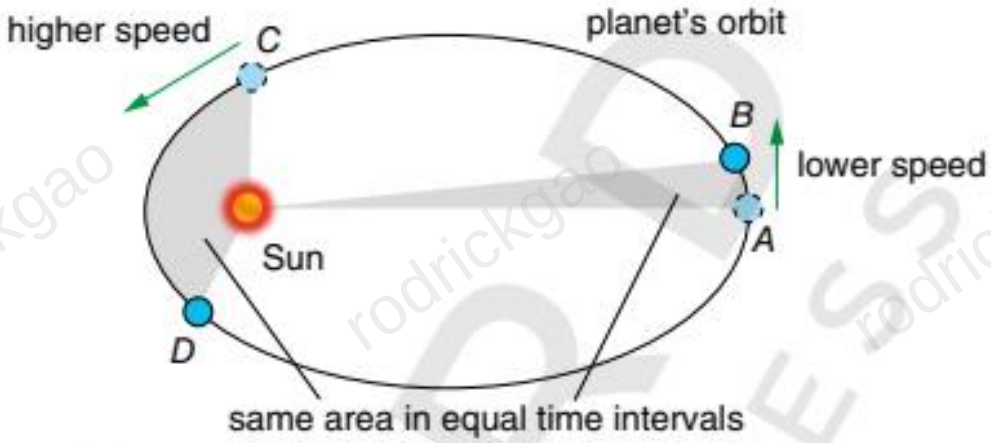


Fig 2.1e *Kepler's second law.*

Equal areas are swept in equal times; orbital speed is greater near perihelion.

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E1.2.2 Kepler's second law

The line joining a planet to the Sun sweeps equal areas in equal times, so the planet moves faster nearer the Sun and slower farther away. The equal quantity is area, not arc length or angular displacement. Consider both distance and velocity direction when comparing perihelion and aphelion speeds.

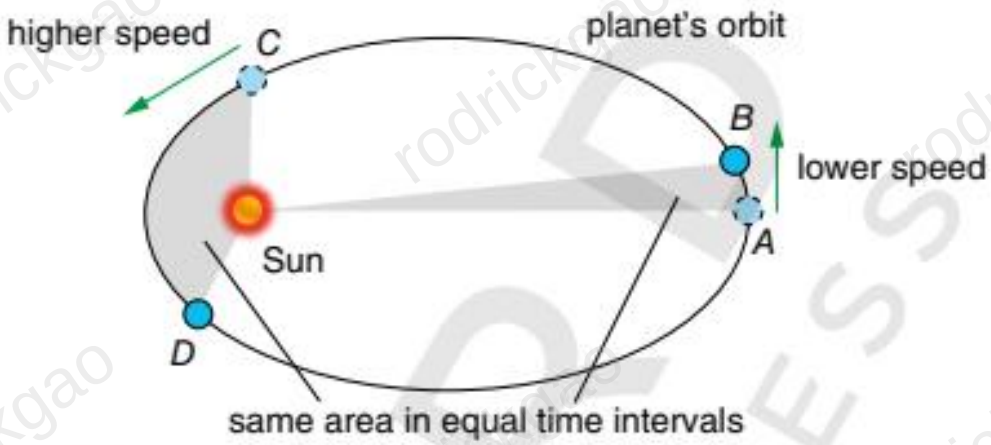


Fig 2.1e *Kepler's second law.*

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E1.2.3 · E1.2.4 Kepler's third law

For bodies orbiting the same dominant central mass, period squared is proportional to semi-major axis cubed. Semi-major axis equals radius only for a circular orbit. A larger central mass gives a shorter period at the same orbital scale. Do not assume the proportionality constant is identical for different central bodies.

$$T^2 \propto a^3$$

$$\frac{T^2}{a^3} = \text{constant}$$

$$T^2 = \frac{4\pi^2}{GM} a^3$$

Kepler's third law for bodies orbiting the same dominant central mass. T: period; a: semi-major axis, not instantaneous distance.

Orbiting mass much smaller than central mass M (kg); T: s; a: semi-major axis (m); G = 6.67 × 10^{-11} N m^2 kg^{-2}.

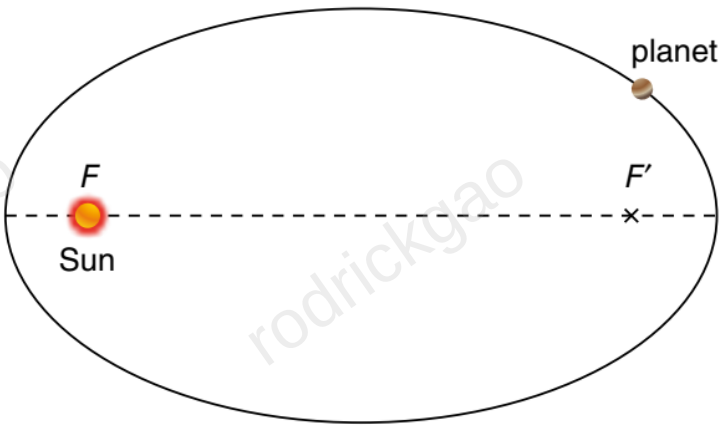


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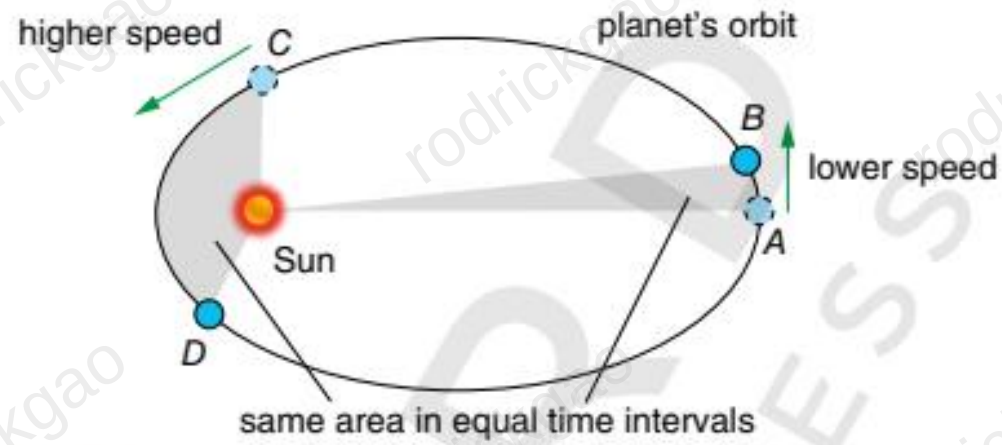


Fig 2.1e *Kepler's second law.*

Equal areas are swept in equal times; orbital speed is greater near perihelion.

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E1.2.5 Apparent weightlessness

An object and its spacecraft falling freely together need no floor support force, producing apparent weightlessness. Gravity has not disappeared: in Earth orbit it supplies centripetal acceleration. The vanishing quantity is support force or the corresponding scale reading, not mass or necessarily gravitational force.

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E1.2.6 Negative gravitational potential energy

Choosing gravitational potential energy to be zero at infinity makes it negative for attracting masses at finite separation. Energy must be supplied to separate them to infinity. Moving farther away raises potential energy towards zero. Whether an orbit is bound depends on total kinetic plus potential energy, not negative potential energy alone.

$$U = -\frac{GMm}{r}$$

U: gravitational potential energy (J), zero at infinite separation; M, m: kg; r: centre separation (m). Point masses or non-overlapping spherical masses. Negative U alone does not imply a bound orbit; total energy must be negative.

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E1.2.7 Energy conservation in orbital motion

When gravity alone does work, kinetic plus gravitational potential energy remains constant. Approaching the central body lowers potential energy and usually increases speed; receding does the reverse. Use the same energy zero for both states. Rocket thrust or resistance requires additional energy transfers to be included.

$$K_1 + U_1 = K_2 + U_2$$

Only gravity does work; K: kinetic energy (J); U: gravitational potential energy (J), with the same zero level in both states.

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E1.2.8 Escape speed

Escape speed is the minimum initial speed at a specified position needed to reach infinity without further thrust, neglecting resistance. In the limiting case, final speed tends to zero at infinity. Measure radius from the body's centre. Escape speed is independent of the escaping mass and exceeds circular-orbit speed at the same radius.

$$v_e = \sqrt{\frac{2GM}{R}}$$

Escape speed v_e (m s^{-1}); M: central mass (kg); R: starting radius from its centre (m). No air resistance or subsequent propulsion; final speed tends to zero at infinity.

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